# Mathematical Writing Template: Structured Draft

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**Abstract.** This document is a reusable template for theorem-driven mathematical writing. It demonstrates a practical workflow for introducing notation, stating assumptions, proving intermediate claims, and organizing appendices and references. Replace placeholders progressively while preserving labels, references, and structural conventions.

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# 1 How to Use This Template

This chapter is intentionally written as mathematically plausible placeholder exposition. The goal is to offer a strong starting structure rather than minimal filler. The recommended workflow is:

1. declare assumptions and notation early,
2. state local lemmas before global statements,
3. keep labels stable and descriptive,
4. move long technical checks to the appendix.

For style guidance, compare the compact statement in Lemma 1.1 with the full proof in Appendix A.1. For citation examples, see [3, 1].

## 1.1 Preliminary Markers

Preliminary markers record background facts or references that should remain visible while drafting. They are useful for tracking external sources for assumptions, conventions, or imported results.

**PRELIM: 1** Confirm that the finite-dimensional norm convention matches the standard source before using compactness. *[3]*

**PRELIM: 2** Check whether the cited gradient-flow reference should be replaced by a domain-specific source.

To print a consolidated preliminary marker list, call:

# Preliminary Marker List

| # | Description | Reference |
|---|---|---|
| 1 | Confirm that the finite-dimensional norm convention matches the standard source before using compactness. | [3] |
| 2 | Check whether the cited gradient-flow reference should be replaced by a domain-specific source. | |

## 1.2 Template TODO Workflow

The template includes an integrated TODO system for in-draft task tracking. Typical usage is to leave short actionable items in context:

**TODO: 1** Replace the placeholder title and author block in main.tex.

**TODO: 2** State theorem assumptions as a numbered list before long proofs.

**TODO: 3** Add one domain-specific running example in the first section.

To print a consolidated task list, call:

# Todo List

| # | Caption |
|---|---|

| | |
|---|---|
| 1 | Replace the placeholder title and author block in main.tex. |
| 2 | State theorem assumptions as a numbered list before long proofs. |
| 3 | Add one domain-specific running example in the first section. |
| 4 | Example of Inline TODO. Note that you must put the slash immediately after the previous word without space to avoid extra space in the output |

## 1.3 Standing Notation

Let $V$ be a finite-dimensional real inner-product space with norm $\|\cdot\|$. We write

$$\langle x, y\rangle \text{ for the inner product} \qquad \mathbf{0} \text{ for the zero vector}$$

and denote the closed unit ball by

$$B := \{x \in V \mid \|x\| \leq 1\}$$

Given a differentiable map $F : V \to \mathbb{R}$, define the gradient flow residual

$$\mathcal{R}_\lambda(x) := x - \lambda \nabla F(x) \qquad \lambda > 0 \tag{1}$$

We will repeatedly reference Equation 1 and the energy functional below:

$$\mathcal{E}(x) := \frac{1}{2}\|x\|^2 + \Phi(x) \tag{2}$$

where $\Phi : V \to \mathbb{R}$ is convex and continuously differentiable.

**Definition 1** Admissible Sequence

A sequence $(x_n)_{n\geq 0}$ in $V$ is *admissible* if there exists $\lambda \in (0, 1]$ such that

$$x_{n+1} = \mathcal{R}_\lambda(x_n)$$

for all $n \geq 0$.

**Remark 1**

In an actual project, replace Definition 1 by your domain-specific object and keep the same pattern: definition, immediate comment, first estimate, then structural theorem.

## 1.4 Sample Local Estimates

▶ **Lemma 1.1** Coercive Bound. *Assume* $\Phi(x) \geq -c_0 + c_1\|x\|$ *for constants* $c_0, c_1 \geq 0$. *Then*

$$\mathcal{E}(x) \geq \frac{1}{2}\|x\|^2 - c_0 + c_1\|x\|$$

*so every sublevel set* $\{x \in V \mid \mathcal{E}(x) \leq M\}$ *is bounded.*

**Proof.** The inequality is immediate from Equation 2. Boundedness of sublevel sets follows because the right-hand side grows quadratically in $\|x\|$. ◀

▶ **Proposition 1.2** One-Step Descent. *Suppose* $\nabla F$ *is* $L$*-Lipschitz and* $0 < \lambda \leq L^{-1}$. *For an admissible sequence,*

$$F(x_{n+1}) \leq F(x_n) - \frac{\lambda}{2}\|\nabla F(x_n)\|^2 \tag{3}$$

**Proof.** Apply the standard smoothness inequality for $F$ at $(x_n, x_{n+1})$ and substitute $x_{n+1} = x_n - \lambda\nabla F(x_n)$. ◀

▶ **Corollary 1.3** Summability of the Gradient. *Under the assumptions of Proposition 1.2,*

$$\sum_{n=0}^{N} \|\nabla F(x_n)\|^2 \leq \frac{2}{\lambda}\big(F(x_0) - \inf_V F\big)$$

*for every* $N \in \mathbb{N}$.

**Proof.** Sum Equation 3 from $n = 0$ to $N$ and telescope. ◀

## 1.5 Sample Main Statement

▶ **Theorem 1.4** Template Convergence Statement. *Let* $(x_n)$ *be admissible. Assume:*

1. $F$ *is bounded below and has* $L$*-Lipschitz gradient,*
2. *every bounded sequence in* $V$ *has a convergent subsequence,*
3. *the set of critical points of* $F$ *is closed.*

*Then every accumulation point of* $(x_n)$ *is critical for* $F$.

**Proof.** By Corollary 1.3, $\|\nabla F(x_n)\| \to 0$ along a subsequence. Boundedness follows from Lemma 1.1, so compactness yields a convergent subsequence $x_{n_k} \to x_*$. Continuity of $\nabla F$ gives $\nabla F(x_*) = 0$. ◀

> **Example 1** Discrete Model Prompt

A common specialization is $V = \mathbb{R}^d$ and

$$F(x) = \sum_{i=1}^{m} \log\big(1 + \exp(-b_i \langle a_i, x \rangle)\big) + \alpha \|x\|_1 \tag{4}$$

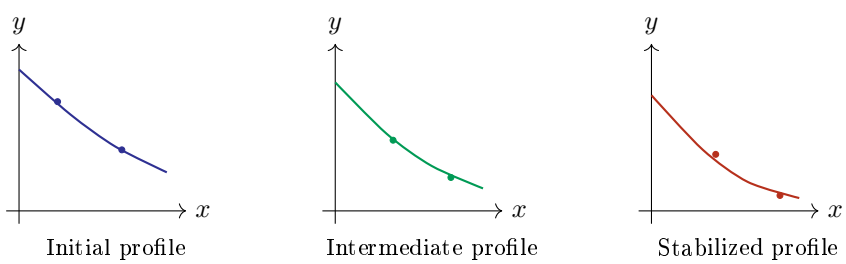
where $(a_i, b_i)$ are data and labels. Replace Equation 4 with your own objective and keep the theorem/proof skeleton intact.

**Remark 2**

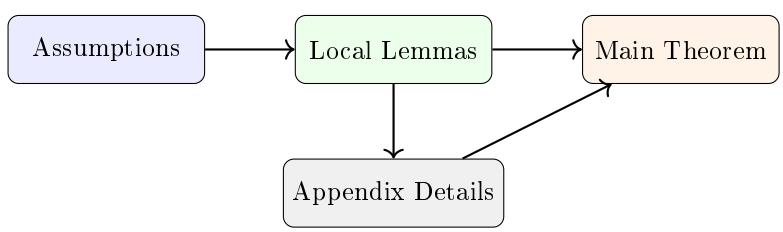
If a proof becomes long, retain the theorem in the main text and move routine verification to Appendix A, as illustrated in Section A.1.

## 1.6 Template Figures and Inputs

This section doubles as a reusable TikZ gallery. The layout keeps the project convention that chapter text includes lightweight wrappers from 'figures/examples/', while each wrapper can reuse modular drawing fragments from 'elements/examples/'.




*Sample multi-panel diagram built from reusable element files.*

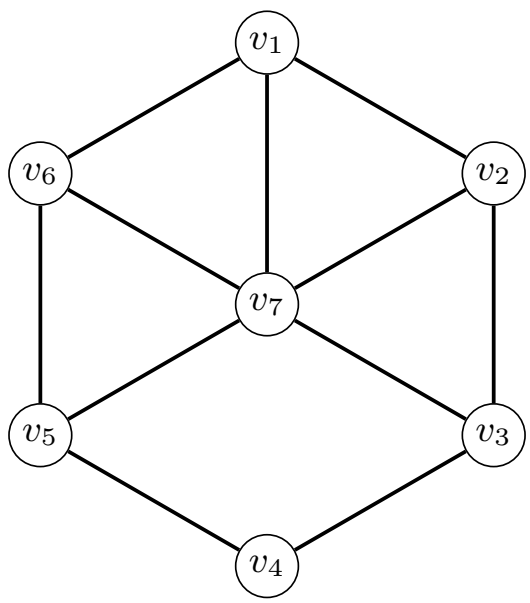



*Template workflow: assumptions to local results to global theorem, with technical details deferred to an appendix.*

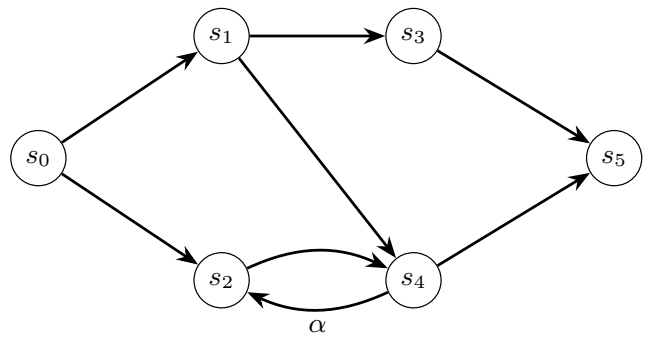
$$
\begin{tikzcd}
A' \arrow[r, "\text{f'}"'] \arrow[rr, bend left, "\text{h'}"] \arrow[d, "\text{u}"'] \arrow[dr, dashed, "\mathscr{S}"] & B' \arrow[r, "\text{g'}"'] \arrow[d, "\text{v}"] & C' \arrow[d, "\text{w}"] \\
A \arrow[r, "\text{f}"] \arrow[rr, bend right, "\text{h}"'] & B \arrow[r, "\text{g}"] & C
\end{tikzcd}
$$

$g \circ f = h,\ g' \circ f' = h',\ v \circ f' = f \circ u,\ w \circ g' = g \circ v.$

*Category-style commutative diagram with aligned morphism families and explicit commutativity relations.*

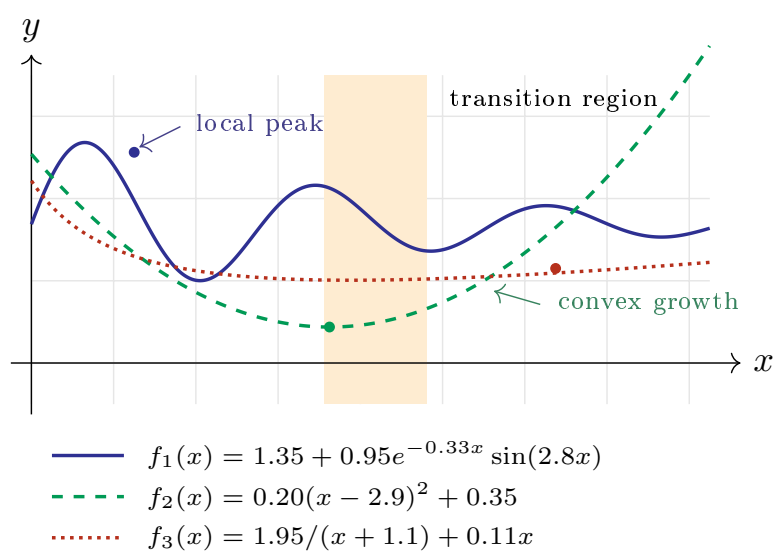



Undirected graph with a symmetric outer cycle and a structured interior hub

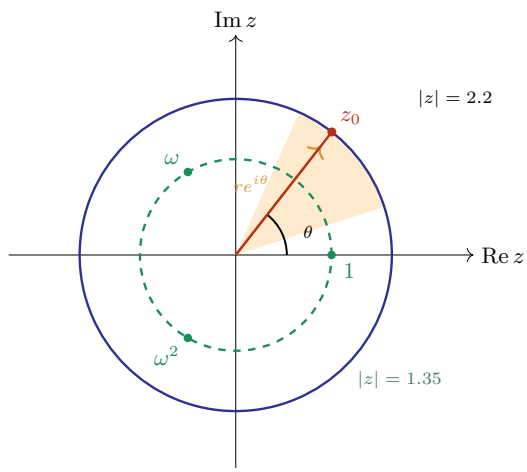



Directed graph with layered flow, branching, and a labeled feedback edge

*Reusable graph templates for combinatorial structure and transition dynamics.*

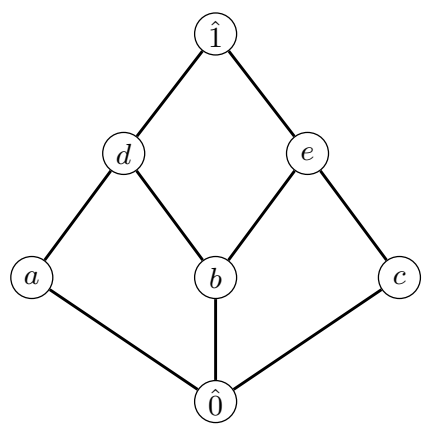



Multi-style $x$-$y$ plot with separated legend, highlighted transition region, and local annotations




Complex-plane sketch with modulus circles, cubic roots of unity, and polar-angle geometry

*Analytic plotting examples for asymptotics, model comparison, and geometric interpretation.*




*Optional Hasse diagram template with noncrossing comparability edges for order-theoretic arguments.*

When adapting these figures, keep style choices consistent: similar line weights, comparable node sizing, and concise labels that avoid collisions at final print scale.

## 1.7 Cross-Reference Checklist

This chapter intentionally demonstrates references to equations Equations 1–3, definition Definition 1, theorem Theorem 1.4, and appendix sections Sections A and B.

For additional writing style examples, see [4, 2].

# A Technical Appendix

This appendix collects technical material that would interrupt the flow of the main narrative.

## A.1 Detailed Proof Placeholder

*Expanded proof of Lemma 1.1.* From Equation 2 and the growth assumption on $\Phi$,

$$\mathcal{E}(x) = \frac{1}{2}\|x\|^2 + \Phi(x) \geq \frac{1}{2}\|x\|^2 - c_0 + c_1\|x\|$$

Now let $M \in \mathbb{R}$ and define

$$\Omega_M := \{x \in V \mid \mathcal{E}(x) \leq M\}$$

If $\|x\| \to \infty$, the right-hand side above tends to $\infty$, so eventually $\mathcal{E}(x) > M$. Hence $\Omega_M$ is bounded. ◀

## A.2 Reusable Lemma Skeleton

▶ **Lemma A.1** Continuity under Composition. *Let $X, Y, Z$ be topological spaces, with $f : X \to Y$ and $g : Y \to Z$ continuous. Then $g \circ f : X \to Z$ is continuous.*

**Proof.** For any open set $U \subseteq Z$, we have

$$(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$$

Since $g$ and $f$ are continuous, $g^{-1}(U)$ is open in $Y$ and then $f^{-1}(g^{-1}(U))$ is open in $X$. ◀

**TODO: 4** Example of Inline TODO. Note that you must put the slash immediately after the previous word without space to avoid extra space in the output

# B Glossary of Notation

| Notation | Description | Reference |
|---|---|---|
| $V$ | Ambient finite-dimensional vector space. | Section 1.3 |
| $F$ | Objective functional in the main argument. | Section 1.3 |
| $\mathcal{R}_\lambda$ | Residual/update map. | Equation 1 |
| $\mathcal{E}$ | Energy functional. | Equation 2 |
| $(x_n)$ | Admissible sequence. | Definition 1 |
| $\Omega_M$ | Template sublevel set used in coercivity arguments. | Section A.1 |

## References


[1] Lawrence C. Evans. *Partial Differential Equations*. 2nd ed. Vol. 19. Graduate Studies in Mathematics. American Mathematical Society, 2010 [Local copy].

[2] Donald E. Knuth. *The TeXbook*. Addison-Wesley, 1984 [Local copy].

[3] Walter Rudin. *Principles of Mathematical Analysis*. 3rd ed. McGraw-Hill, 1976 [Local copy].

[4] Terence Tao. *An Introduction to Measure Theory*. American Mathematical Society, 2011 [Local copy].